SpaceX Falcon 9

First Stage landing prediction



Presented by: **GoGitter001**

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion





Executive Summary

- **Core Goal**: Predict the landing success of SpaceX Falcon 9 first-stage boosters.
- **Business Value**: Lower launch costs due to first stage reuse.
- **Impact**: Enables upcoming competitor SpaceY to accurately estimate costs and bid competitively for launch contracts.

Data and analytics

- Data Sources: Extracted 90 clean records using SpaceX REST API and BeautifulSoup web scraping.
- Exploratory Data Analysis: Run via SQL (SQLite) queries and Seaborn/Matplotlib charts.
- Geospatial & Dashboards: Built interactive dashboards via Plotly Dash and mapped sites with Folium.

Key insights

- Launch Sites: KSC LC-39A holds the highest success rate; CCAFS LC-40 improved rapidly as the rocket version matured.
- Orbit Impact: SSO, HEO, and LEO orbits yield the highest first-stage recovery rates.
- Payload Mass: Success rates scale positively with heavier payloads up to 8,000 kg.

Predictive analytics with ML

- Tested on an 18-record sample dataset (20% split).
- Used Decision Tree, Support Vector Machine, Logistic Regression and KNN.
- All 4 models gave an accuracy of ~83.3%

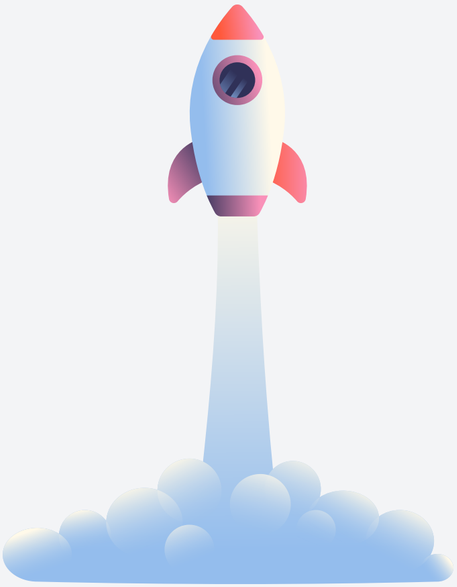
Conclusion

- Deploy the Decision Tree Classifier for predictive deployment due to its superior training optimization (87.14%).
- Maximize booster recovery, target SSO/LEO orbits, launch from KSC LC-39A, and carry payload weights between 3,000 kg and 7,000 kg.

Introduction

The objective of this project is to ascetrain the viability of a new space company SpaceY to compete against SpaceX. Here we want to arrive at:

- The estimated total cost of launches and success rates
- The best places to launch from



Methodology

Data Collection

- Pulled core launch data via the SpaceX REST API using Python.
- Extracted historical launch tables from Wikipedia using BeautifulSoup.
- Created a binary target variable for launches (1 = Success, 0 = Failure).

Data wrangling

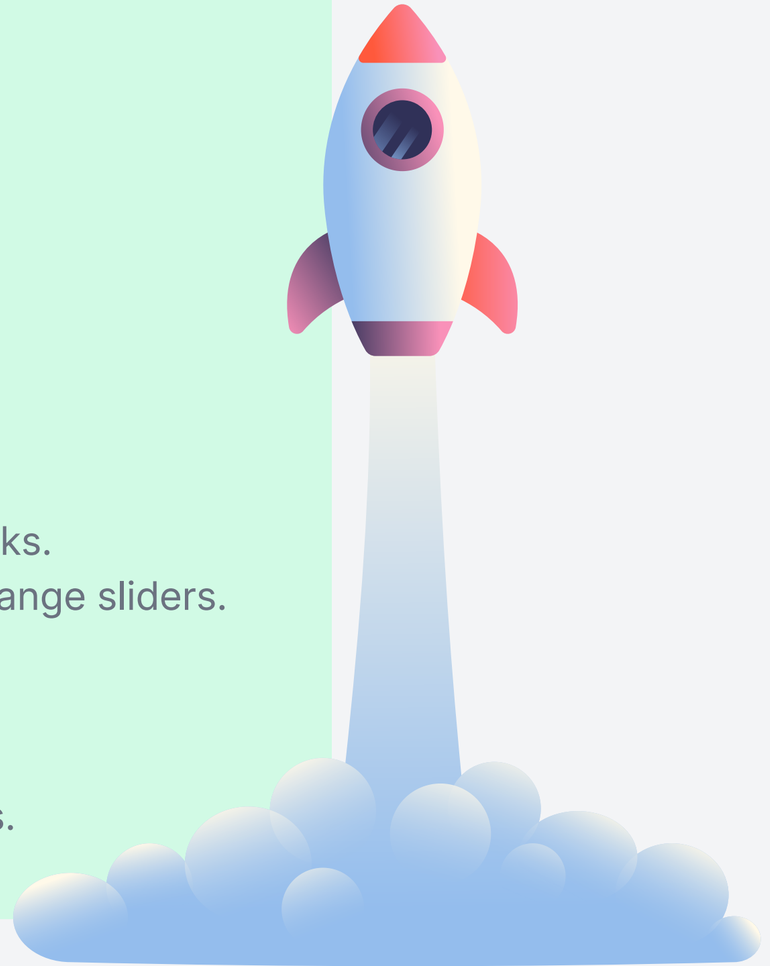
- Isolated Falcon 9 flights exclusively (90 total records).
- Filled missing payload values using column averages.
- Transformed categorical fields (LaunchSite, Orbit) using One-Hot Encoding.

Exploratory Analysis (EDA) & Visualisations

- Used SQL to analyze metrics, landing frequencies, & payload totals.
- Built Seaborn and Matplotlib charts to isolate performance trends.
- Folium: Mapped launch pads to analyze proximity to coastlines and transport links.
- Plotly Dash: Developed an interactive dashboard with dropdowns and payload range sliders.

Predictive Analytics using ML

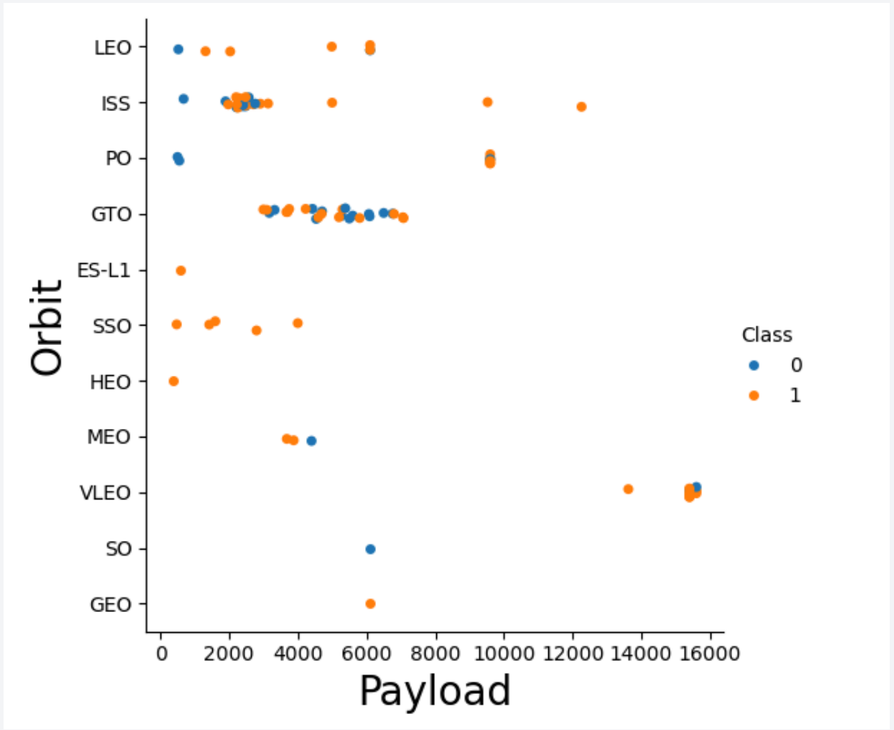
- Divided dataset into an 80/20 train/test split (18 test records).
- Used GridSearchCV to tune Logistic Regression, SVM, KNN, and Decision Trees.
- Evaluated models using accuracy scores and confusion matrices.



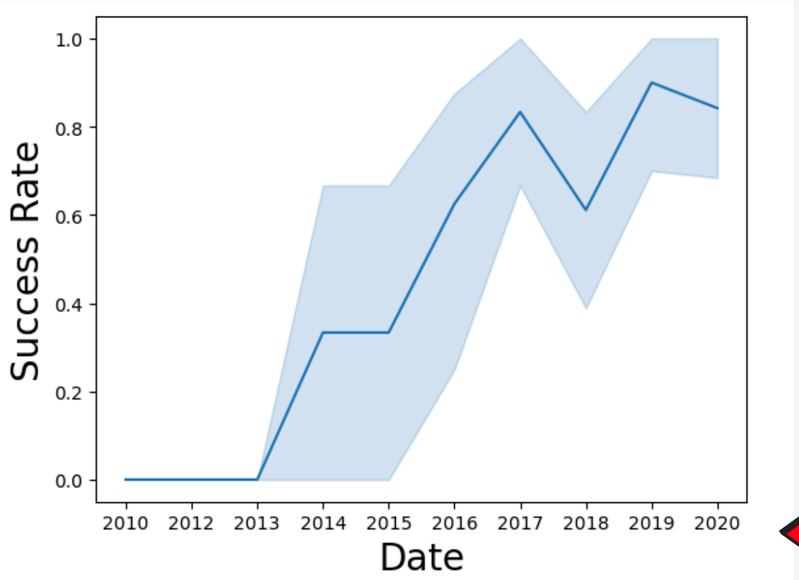
RESULTS

Results of Exploratory Data Analysis (EDA)

- SpaceX uses 4 different launch sites
- The average payload of Falcon 9 v1.1 is 2928 kg
- The first successful landing happened in 2015, five years after the first launch
- Two booster versions failed at landing on drone ships in 2015: F9 v1.1 B1012 and F9 v1.1 B1015
- The number of landing outcomes improved as years passed



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.



You can observe that the sucess rate since 2013 kept increasing till 2020

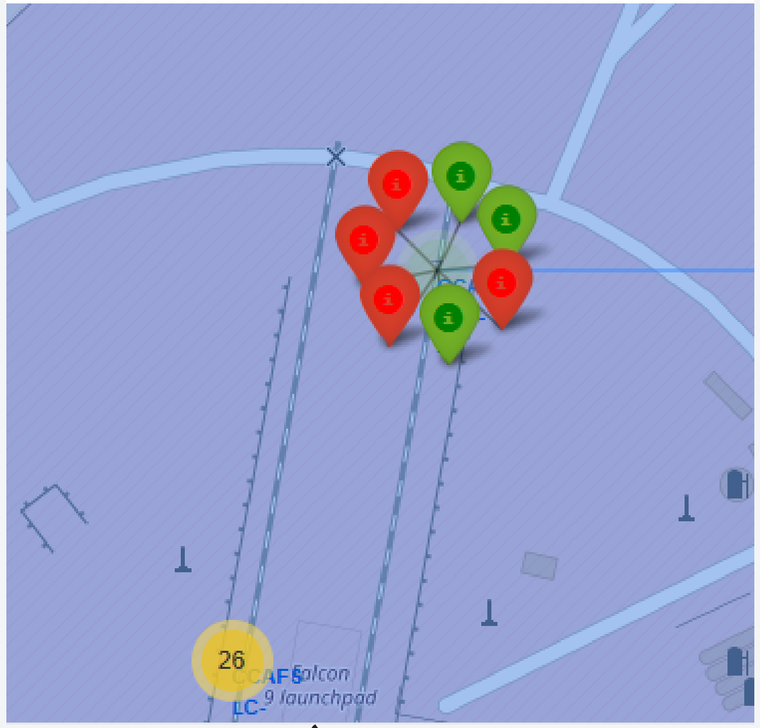


RESULTS

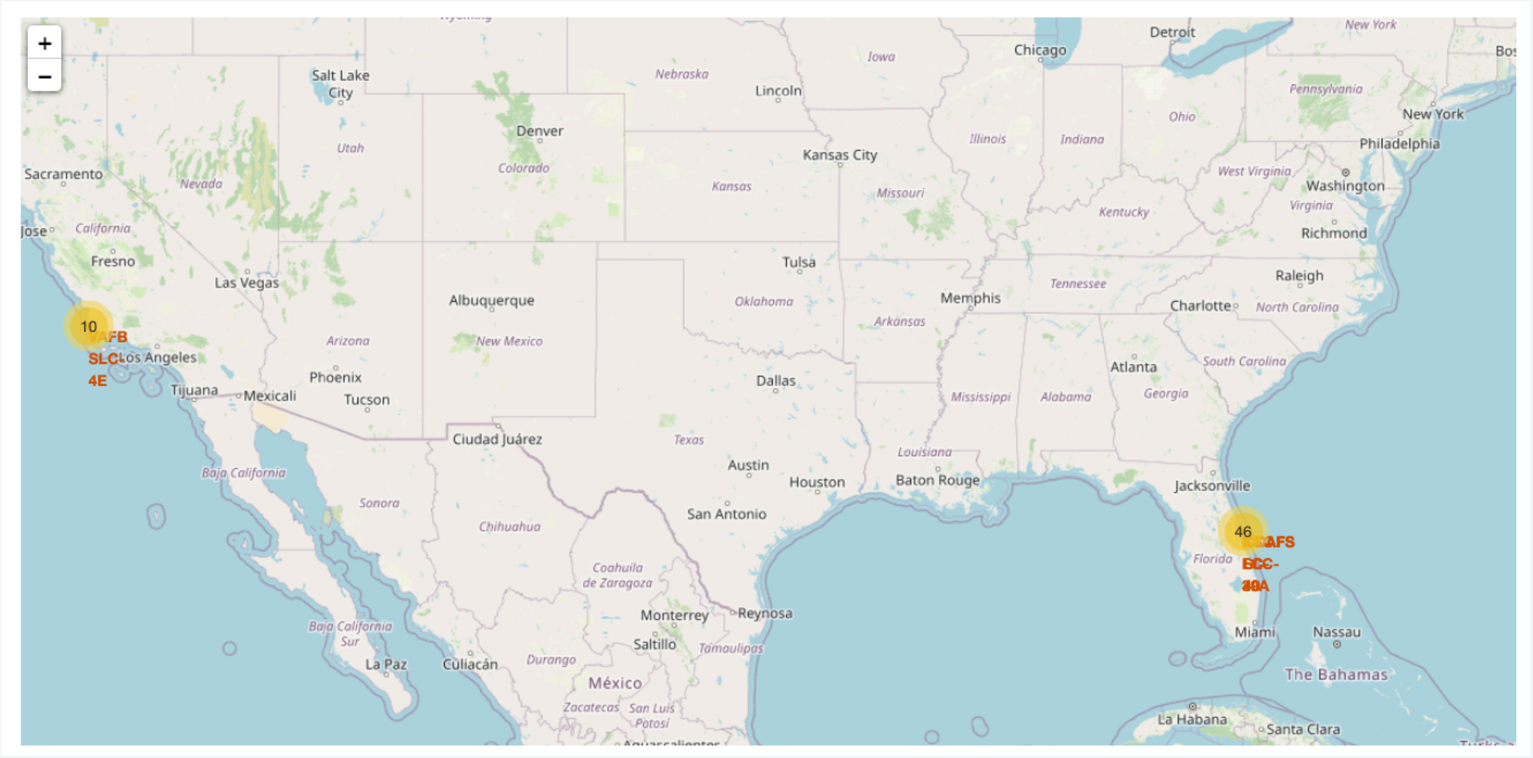


EDA using Visualisations

- Launch sites are usually on the coasts for safety reasons
- Most launches happened from the East Coast



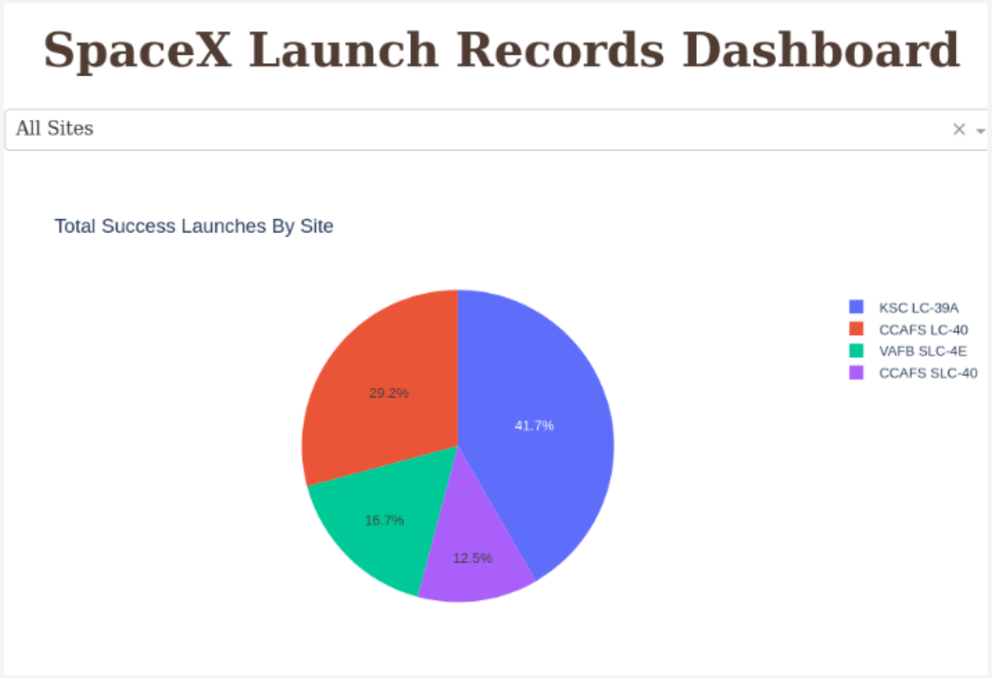
Launch site with outcomes

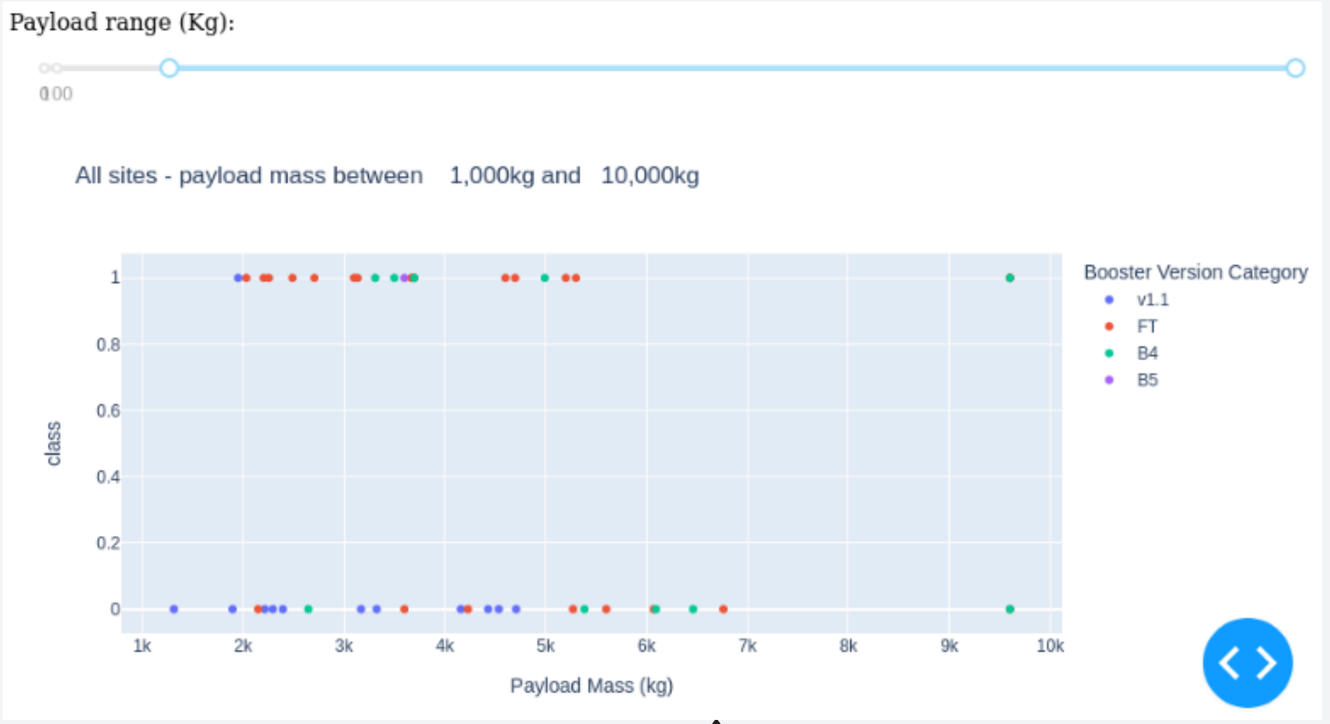


RESULTS

Dashboards







- The place of launch seems to be an important factor for success.
- Payloads of under 6000kg and FT boosters are the most successful

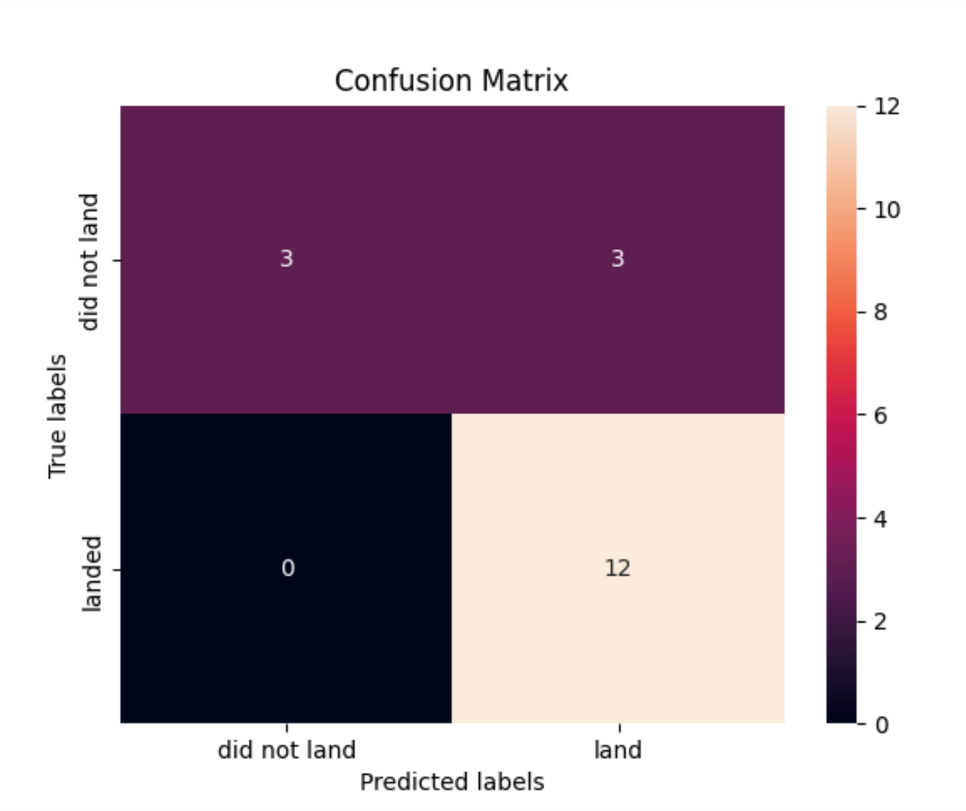
Payload vs. Launch Outcome

RESULTS

Predictive analytics using Machine Learning

- All four models proved highly reliable with a tied 83.33% accuracy on unseen test data, successfully mapping the core features that dictate booster recovery.
- The Decision Tree Classifier achieved the highest overall training accuracy optimization
- Confusion matrix of Decision Tree Classifier proves its accuracy by showing higher true positives and true negatives compared to the false ones.





All 4 models gave the same confusion matrix

Conclusion

- Data was obtained from different sources (spacex API, wikipedia)
- Successful landing outcomes improve over time
- Launches above 7000kgs are riskier
- The best launch site is KSC LC-39A
- A decision tree classifier can be used to predict successful landings which will be useful for the upcoming company SpaceY to plan its launches



Thank You



Presented by: **GoGitter001**